

# Aditya Chowdhury

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Portfolio: <https://adichowdhury15.github.io/>

## EDUCATION

### Northeastern University, College of Engineering

**Boston, MA**

*Bachelor of Science in Mechanical Engineering and Design*

*Expected Graduation December 2027*

**Relevant Coursework:** Cornerstone of Engineering I-II, Statics, Thermodynamics, Mechanics of Materials, Fluid Mechanics, Material Science, Differential Equations and Linear Algebra

**Clubs & Activities:** American Society of Mechanical Engineers (ASME), RISE Against Hunger

## PROJECT WORK

### University Registration System

*Aug. 2026*

- Designed and implemented a Python-based course registration system with nested data structures.
- Enforced a 13-credit limit per student, seat capacity checks and duplicate-enrollment prevention.
- Built a modular, function-based architecture with input validation and a full test suite.

### Full Scale PC Model

*July 2025 - Ongoing*

- Independently modeled a full-scale PC in SolidWorks, producing precise replicas of the front, back and top panels.
- Hand-measured all components with a caliper and applied GD&T standards for consistent dimensional accuracy.
- Diagnosed and resolved complex part-integration issues through iterative refinement for a seamless assembly.

### Environmental Sustainability Exhibit

*Feb. 2025 - May 2025*

- Engineered an interactive children's museum exhibit teaching families about energy use and renewable power.
- Integrated Arduino sensing, resistor-encoded 3D-printed buildings, and MATLAB for real-time RGB LED feedback.
- Designed a durable, child-friendly three-point contact system for reliable, error-free voltage detection.

### Assistive Projectile Launcher

*Jan. 2025 - Feb. 2025*

- Collaborated with a team to build a one-handed orthotic for users with mobility and motor-control limitations.
- Applied the engineering design process across prototypes and testing iterations, maintaining an accurate BOM.
- Delivered a low-cost design within strict constraints: \$10 budget, 2.5 lb weight limit, defined size restrictions.
- Achieved a top-three finish in the class design and shooting competition with a consistent, reliable device.

## WORK EXPERIENCE

### Panera Bread Bakery

**Freehold, NJ**

*Dining Room Caretaker and Dishwasher*

*Nov. 2022 - Aug. 2023*

- Maintained dining room cleanliness and restocked supplies for a comfortable customer experience.
- Coordinated with the line team to keep a steady flow of clean dishes to the kitchen and dining area.

### Domino's Pizza

**Freehold, NJ**

*Line Cook and Cashier*

*Mar. 2022 - Sept. 2022*

- Prepared pizzas and a variety of menu items in a high-stress environment, prioritizing quality.
- Took orders and assisted customers at the front counter during peak hours with reliable service.

## SKILLS & TOOLS

**CAD & Simulation:** SolidWorks, AutoCAD, Rhinoceros 3D, Technical Drawings, Assemblies

**Manufacturing & Design:** GD&T, Tolerance Stack-Up, DFA/DFM, Dimensional Inspection, Mechanism Design

**Testing & Instrumentation:** DAQ, V&V, Root-Cause Analysis, Wiring Diagrams, Pressure/Temp Sensors

**Programming & Controls:** MATLAB, Python, Arduino, Java, Dr. Racket